

Alessandro Salo

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I am a Game & Media Technology MSc student at Utrecht University specialising in graphics programming, GPGPU, and optimization. I have a strong foundation in mathematics and physics, with proven ability to adapt and excel in computer science. Aiming to build the technologies for artists and developers to create beautiful things.

TECHNICAL SKILLS

Specializations	Graphics Programming, Performance Optimization, GPGPU, Physics, Mathematics, Machine Learning, Game Development, Algorithms
Programming Languages	C/C++ (4+ yrs), Python (5+ yrs), C# (2+ yrs), Bash (4+ yrs), HTML, CSS, JavaScript
Technologies & APIs	Git, CMake, Linux, Profilers, OpenCL, Vulkan, OpenGL, GLSL, Unity, PyTorch, NumPy, OpenCV

EDUCATION

MSc Game and Media Technology — Utrecht University 2025 – present
GPA: 8.0/10

MSc Computational Science & Engineering — EPFL 2024 – 2025
GPA: 5.1/6 — *Transferred to Utrecht University*

BSc Physics & Mathematics (Honours) — Utrecht University / UCU 2020 – 2024
GPA: 3.87/4 & 8.6/10

RELEVANT PROJECTS / EXPERIENCE

Ray Tracer with GPU-Parallel LBVH Raytracer-LBVH | BVH
C++, OpenCL, Vulkan *Oct 2025 — ongoing*

- Developed a CPU ray tracer based on RaytracingInOneWeek; extended to support triangle meshes.
- Implemented a CPU-based Bounding Volume Hierarchy (BVH) to accelerate rendering.
- Engineered a fully GPU LBVH using Morton codes, radix sort, and Karras (2012) hierarchy construction using OpenCL, achieving an 89.11× speed-up over the CPU baseline on scenes with 28M triangles.
- Porting to Vulkan for GPU ray tracing and LBVH construction.

Monte Carlo App MonteCarlo
C++ *Nov 2024 — Dec 2024*

- Built an interactive C++ framework for Monte Carlo sampling, statistical analysis, and empirical verification of mathematical properties such as the Central Limit Theorem.
- Designed and implemented an abstract syntax tree (AST) parser to convert user-written mathematical expressions into GUI elements.

ADDITIONAL PROJECTS

- **YOLOv1 Object Detector** (Python, OpenCV)
- **Voxel Reconstruction** (Python, OpenCV)
- **Unnamed Horror Game** (Unity, C#)
- **Unnamed VR Game** (Unity, C#)
- **Between These Walls** (Unity, C#)
- **Hard Sphere Simulations** (C)

EXPERIENCE

Science Student Assistant — Utrecht University, Netherlands Aug 2022 — Jun 2024

- Sole student tutor and TA for all UCU mathematics and physics courses over two years; graded assignments and provided constructive feedback.

RESEARCH PUBLICATIONS

- R. M. Alkemade, **A. Salo**, L. Filion, F. Smallenburg, "*Determining fluid-crystal phase boundaries for a binary hard-sphere mixture using direct-coexistence simulations*," *J. Chem. Phys.*, vol. 164, no. 12, Mar. 2026. doi:10.1063/5.0321591

ACHIEVEMENTS & INTERESTS

Academic	**9.1/10** Advanced Graphics (UU) • **Magna cum laude** UCU • **Cum laude** Utrecht University
Hobbies	Bouldering • Board Games • Hiking • Photography • Music
Interests	Graphics/Rendering • Game Engines • Game Development • Philosophy • Literature